MILITARY SPECIFICATION

CONTROL, PRESSURE, FILLING, NONVENTED DRUM

This specification is approved for use within the USA Relvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a pressure shutoff control (valve) with hose assembly used for filling military standard collapsible drums of 500-gallon capacity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4930

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MILITARY

MIL-S-7916	- Sealing Compound, Thread and Gasket, Fuel, Oil, and Water Resistant.
MIL-F-46162	- Fuel, Diesel, Referee.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria For Military Systems, Equipment and Facilities.

2.1.2 Other Government drawings. The following other Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA13215E8372	- Control Assembly, Pressure Filling.
D13217E2820	- Container, Shipping, Reusable.

(Copies of specifications, standards, and drawings, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The control, pressure, filling, nonvented drum (herein after called “control assembly”) shall be in accordance with Top Assembly TA13215E8372 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fit, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit,

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assembly, and operation of the control assembly. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures,etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of the control assembly from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish control assemblies conforming to this specification.

3.3 Material. Material shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The control assembly shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the control assembly may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponents, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Sealing compound. Sealing compound conforming to MIL-S-7916 shall be used on all pipe threads.

3.4 Environmental conditions.

3.4.1 Operating temperature. The control assembly shall perform as specified in ambient temperatures from -40 +3 °F to +125 +3 °F.

3.5 Functional use. The control assembly shall perform the following functions without leaking or deformation:

a. Manually start the flow of fluid through the control assembly when the fill button of the cyclic valve is depressed.
b. Manually stop the flow of fluid through the control assembly when the stop button of the cyclic valve is depressed.
c. Automatically stop the flow of fluid through the control assembly when pressure at the venturi throat reaches 5, ±.05 pounds per square inch gauge (psig).

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3.6 Hydrostatic pressure. The assembled inlet valve body, outlet valve body, and valve cover shall withstand a hydrostatic pressure of 200 psig for 2 minutes without leaking, sweating, or visible deformation at any point on the surface.

3.7 Identification marking. The control assembly shall be identified in accordance with MIL-STD-130.

3.8 Workmanship. The control assembly shall be free from any defects which could reduce serviceability. The castings shall be clean, sound, and free from blowholes, hardspots, porosity, cracks, and other injurious defects. All burrs and sharp corners shall be removed.

3.9 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the control assembly may be newly fabricated from recovered materials to the maximum extent practicable, provided the control assembly produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the control assembly.

3.10 Human factors engineering. The control assembly shall conform to human engineering criteria as described in MIL-STD-1472, paragraphs 4 (General Requirements), 5.4 (Controls), 5.5 (Labeling), 5.6 (Anthropometry), 5.9 (Design in Maintainability), and 5.13 (Hazards and Safety).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards as applicable.

4.1.2 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspections requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4)
c. Inspection of packaging (see 4.6)

4.3 First article inspection.

4.3.1 Examination. The first article control assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article control assembly shall be tested as specified in table I. Failure of any test shall be cause for rejection.

TABLE I. First article tests.

First Article	Individual	Test	Requirement paragraph	Test paragraph
X		Valve Hydrdrostatic pressure.	3.6	4.5.2.1
		Control Assembly.		
X	X	Cycle Assembly.	3.5	4.5.2.2
X	X	Leakage.	3.6	4.5.2.3
X	X	Function.	3.5	4.5.2.4

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL (Acceptable Quality Level) shall be 1.0 percent defective.

4.4.3 Tests.

4.4.3.1 Valve. Each assembled inlet valve body, outlet valve body, and valve cover shall be tested as specified in table I. Failure of the test shall be cause for rejection.

4.4.3.2 Control assembly. Each control assembly shall be tested as specified in table 1.

4.5 Inspection procedure.

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4.5.1 Examination. The control assembly shall he examined as specified herein for the following major defects:

101. Material not as specified.
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes, or treatment.
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the control assembly.
106. Pipe threads not coated as specified.
107. Sealing compound not as specified.
108. Workmanship not as specified.
109. Identification marking missing, incomplete, or illegible.
110. Human factors engineering not as specified.

4.5.2 Tests.

4.5.2.1 Hydrostatic. Subject the assembled inlet valve body, outlet valve body, and the valve cover to a hydrostatic pressure of 200 psig for a period of not less than 2 minutes. Nonconformance to 3.6 shall constitute failure of this test.

4.5.2.2 Test cycle. Using test fluid conforming to MIL-F-46162, one testing cycle shall consist of (1) depressing the fill button of the cyclic valve to start the flow of fluid, (2) depressing the stop button of the cyclic valve to stop the flow of fluid and, (3) depressing the fill button of the cyclic valve to start the flow of fluid and allowing the fluid to flow until automatic shutoff of the flow when the pressure at the venturi throat gauge reads 5 +0.50 psig. During each cycle, in the interval between actuating fill button (start of flow) and automatic shutoff, a minimum of three readings of the pressure shall be taken simultaneously at the drum gauge and venturi throat gauge and shall be recorded.

4.5.2.3 Leakage test. Each control assembly shall be tested for leakage. The valve inlet shall be assembled to a pumping unit and the valve exit hose assembly shall be assembled to OPW fitting No. 77A and No. 7713 or equal. The OPW fittings shall be assembled to a single opening drum having provisions for a pressure gauge connection and a drainage connection. When fluid flow through the control assembly has been initiated, the stop button of the cyclic valve shall be depressed to stop the flow of fluid through the valve. The static line pressure at the inlet side of the valve shall be less than 60 psig. The brass plug on the underside of the valve at the venturi throat section shall be removed. The valve shall be in the closed position for 15 minutes, +30 seconds, and shall be inspected during this interval for leaks at the brass plug opening and valve surface. Any evidence of continuous dripping shall constitute failure of this test. If there is no evidence of leakage, the brass plug shall be inserted in the plug opening at the venturi throat section and the OPW fittings

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and the drainage valve on the drum shall be fully opened. Flow through the valve shall be initiated by depressing the fill button of the cyclic valve. Drainage from the drum shall be such that the pressure reading on the drum gauge shall be not more than 2 psig. An alternate method is to direct the fluid flow from the valve into an open drum. The flow through the valve shall not be less than 50 gallons per minute (gpm) operating at an inlet line pressure of not more than 60 psig (no-flow pressure). The valve shall operate in the open position for a period of 15 minutes, ±30 seconds. Valve closure during this interval shall constitute failure of this test.

4.5.2.4 Functional test. The test equipment shall be as specified in 4.5.2.3. Pressure gauges shall be inserted in the brass plug opening at the venturi throat section of the valve and at the drum gauge connection. The drainage valve on the drum shall be closed and the OPW fittings shall be fully opened. The following sequence of operations shall be performed for 1,000 cycles on the first article sample and for 25 cycles on each valve thereafter: (a) depress fill button of the cyclic valve to start flow, (b) depress stop button of the cyclic valve to stop flow, (c) depress fill button of the cyclic valve to start flow, and (d) automatic shutoff of the flow when the pressure at the venturi throat gauge reads 5 ±0.50 psig. The no-flow operating inlet line ±0.50 pressure shall be not more than 60 psig. The pressure at the drum gauge and venturi throat gauge shall be recorded simultaneously at different intervals during flow. The flow rate through the valve for the first article sample shall be between 20 and 100 gpm at 20 gpm increments for 200 cycles each increment (80 cycles at -40 ±1 °F, and 120 cycles at +125 ±1 °F). The flow rate through each valve tested thereafter shall be not less than 50 gpm at ambient room temperature. The capacity of the drum shall be such that the time interval between depressing the fill button and automatic shutoff is 2 minutes, ±30 seconds. Failure to function properly shall constitute failure of this test. Leakage test as specified in 4.5.2.3 shall be repeated for the first article sample.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

111.	Dust caps and plugs not installed as specified.
112.	Container not in accordance with referenced drawing for level A.
113.	Hose and valve not placed in container as specified for level A.
114.	Strapping not as specified for level A.
115.	Marking illegible, incorrect, or incomplete

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5. PACKAGING

5.1 Packing. Packing shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The furnished dust caps and plugs shall be secured in place on the valve and hose. Each valve and hose shall be packed in a shipping container conforming to PPP-B-601, overseas type, style A, modified as shown on D13217E2820. Each valve shall be placed in the shipping container resting on the saddles. The hose shall be coiled around the interior surfaces of the shipping container and rested on the saddles. The shipping container shall be closed and secured with nine No. 8, 1.250-inch-long flat head wood screws. Additionally, the box shall be strapped in accordance with the appendix to the box specification except that strapping shall be flat and the finish "B".

5.1.2 Level C. Each valve and hose shall be packed for level C as specified in 5.1.1 for level A, except that the box shall be domestic type and the strapping may be finish "A".

5.2 Marking. In addition to any special or identification marking required (see 6.2), each container shall be marked in accordance with MIL-STD-129 and drawing 13217E2880.

6. NOTES

6.1 Intended use. The control assembly will be used in an integrated filling system for shutting off the flow of fluid into military standard collapsible drums (500-gallon capacity) at a predetermined drum pressure of 5 ±0.5 psig.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date this specification.
b. When a first article is required (see 3.2).
c. Level of packing required (see 5.1).
d. Any special markings (see 5.2).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Preservation. For the purpose of marking only, preservation shall be shown as level A.

6.5 Subject term (key word) listing.

Blivet
By-pass
Control
Drum
Filling Control
Fuel Drum
Pressure
Valve

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6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - ME
Air Force - 69

Review activities:

Army - GL
Air Force - 71, 99

User activity:

Navy - MC

Preparing activity:

Army - ME

Project 4930-A261

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-43044C(ME)	Control, Pressure, Filling, Nonvented Drum

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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